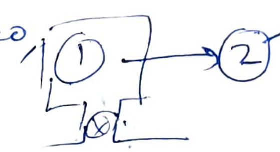
# Steady & Unsteady flow numericals -

**Q.1** An evacuated bottle of $0.5 m^3$ volume is slowly filled from atmospheric air at 1.01325 bar until the pressure inside the bottle also becomes 1.01325 bar. Due to heat transfer, the temperature of air inside the bottle after filling is equal to atmospheric air temperature. Determine the amount of heat transfer.

Soln.

State 1

$V = 0.5 m^3$, $p = 0$,
$m_1, u_1$



$V = 0.5 m^3$, $T = T_{atm}$
$p = 1.01325$ bar
$m_2, u_2$

$\Rightarrow Q + m_i h_i = \cancel{W}^0 + \cancel{m_e h_e}^0 + m_2 u_2 - \cancel{m_1 u_1}^0$

$\Rightarrow Q_{1-2} = m_2 u_2 - m_i h_i \qquad m_i - \cancel{m_e}^0 = m_2 - \cancel{m_1}^0$

$\Rightarrow Q_{1-2} = m_2 u_2 - m_2 \{u_2 + p_a v_a\} \Rightarrow \qquad m_i = m_2$

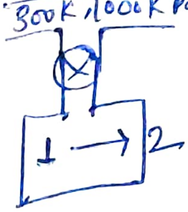
$Q_{1-2} = -m_2 p_a v_a = -p_a V_a = -1.01325 \times 10^2 \times 0.5$
$= -50.67 kJ$

**Q (2)** A $3 m^3$ insulated tank having initial charge of helium at 300 K & 100 kPa is charged from a constant temperature supply line at 300 K and 1000 kPa. If the final tank pressure is 900 kPa, determine the final helium temperature. Take R for helium 2.077 kJ/kg & $\gamma = 1.666$

Soln.

gas - $m = \frac{PV}{RT} - \frac{\frac{N}{m^2} \times m^3}{\frac{kJ}{kg\,K} \times K} = \frac{N\text{-}m}{J} \times kg = \frac{N\text{-}m}{N\text{-}m} \times kg$
$= kg$

300 K, 1000 kPa



at 1 $V_1 = 3 m^3 = V_2$
$P_1 = 100 kPa \rightarrow P_2$ 900 kPa

$$m_i - \cancel{m_e}^{0} = m_2 - m_1 \Rightarrow m_i = m_2 - m_1$$

· Energy Balance → $m_i h_i + \cancel{Q}^{0} = \cancel{W}^{0} + \cancel{m_e h_e} + m_2 u_2 - m_1 u_1$

$$\Rightarrow (m_2 - m_1) h_i = m_2 u_2 - m_1 u_1$$

$$\Rightarrow m_2 h_i - m_1 h_i = m_2 u_2 - m_1 u_1$$

$$\Rightarrow m_2 c_p T_i - m_1 c_p T_i = m_2 c_v T_2 - m_1 c_v T_1$$

$$\Rightarrow m_2 \gamma T_i - m_1 \gamma T_i = m_2 T_2 - m_1 T_1$$

$$\Rightarrow \frac{P_2 V_2}{R T_2} \gamma T_i - \frac{P_1 V_1}{R T_1} \gamma T_i = \frac{P_2 V_2}{R T_2} T_2 - \frac{P_1 V_1}{R T_1} T_1$$

$$\Rightarrow \frac{P_2 \gamma T_i}{T_2} - \frac{P_1 \gamma T_i}{T_1} = P_2 - P_1$$

$$\Rightarrow \frac{900 \times 1.666 \times 300}{T_2} - \frac{100 \times 1.666 \times 300}{300} = 900 - 100$$

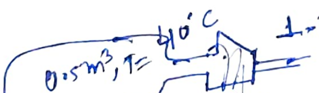
$$\Rightarrow \frac{449820}{T_2} = 800 + 166.6$$

$$\Rightarrow T_2 = 465.36\ K$$

**Emptying of a vessel –**

Q. A compressed air bottle containing $0.5 m^3$ of air at 30 bar and 40°C is used to drive a turbine. The turbine exhausts to atmosphere at 1.013 bar. Determine the maximum amount of work that would be delivered by the turbine.

Sol^n.




$$\cancel{m_i}^{0} h_i + \cancel{Q}^{0} = m_e h_e + W + m_2 u_2 - m_1 u_1$$

$$W = m_1 u_1 - m_2 u_2 - m_e h_e$$

$$\cancel{m_i}^{0} - m_e = m_2 - m_1 \Rightarrow m_e = m_1 - m_2$$

$$\Rightarrow W = m_1 u_1 - m_2 u_2 - (m_1 - m_2) h_e$$

$$m_1 = \frac{P_1 V_1}{R T_1} = \frac{30 \times 10^2 \times 0.5}{0.287\ kJ/kgK \times 313} = 16.698\ kg$$

$T_1 = 40°C = 313 K$

$$m_2 = \frac{P_2 V_2}{R T_2} = \frac{1.013 \times 100 \times 0.5}{0.287 \times T_2} = \frac{176.48}{T_2}\ kg\ K$$

1 → 2 adiabatic process

$$\left(\frac{P_2}{P_1}\right)^{\gamma-1/\gamma} = \frac{T_2}{T_1} \Rightarrow \left(\frac{1.013}{30}\right)^{.4/1.4} = \frac{T_2}{T_1 = 313\,K}$$

$\Rightarrow T_2 = 119K$

$m_2 = 176.48/119 = 1.483\,kg$

$W = m_1u_1 - m_2u_2 - (m_1 - m_2)h_e$

$\downarrow$

$C_pT_e = C_pT_2$

$W = 16.698 \times C_vT_1 - 1.483 \times C_vT_2 - (16.698 - 1.483) \times C_pT_2$

$C_v = 0.718, \quad C_p = 1.005$

$T_1 = 313\,K$

$T_2 = 119\,K$

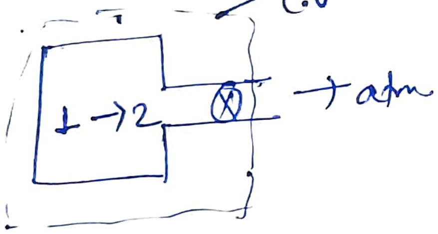
$W = 1806.25\,kJ.$

Q. An air receiver of volume $5.5 m^3$ contains air at 16 bar and 42°c. A valve is opened and some air is allowed to blow out to the atmosphere. The pressure of air in the reciever drops rapidly to 12 bar when the valve is closed. Calculate the mass of air which has left the reciever.

Sol^n

1→2 ⇒ Rev. Adiabatic

C.V




mass balance –

$\cancel{m_i}^{0} - m_e = m_2 - m_1$

$\Rightarrow m_e = m_1 - m_2$ —(1)

Energy balance →

$\cancel{m_i h_i} + \cancel{Q}^{0} = m_e h_e + \cancel{W_r}^{0} + m_2u_2 - m_1u_1$

$P_1 = 16\,bar$

$T_1 = 42°c + 273 = 315\,K$

$V_1 = 5.5 m^3$

$P_2 = 12\,bar$

$T_2 = ?$

$V_2 = 5.5 m^3.$

$\Rightarrow m_e h_e = m_1u_1 - m_2u_2$

$(m_1 - m_2)h_e = m_1u_1 - m_2u_2$

mass of air left →

$m_e = m_1 - m_2$

$$m_1 = \frac{P_1V_1}{RT_1} = \frac{16 \times 10^2\,N/m^2 \times 5.5\,m^3}{0.287\,\frac{kJ}{kgK} \times 315\,K} = 97.34\,kg$$

$$m_2 = \frac{P_2V_2}{RT_2} = \frac{12 \times 10^2 \times 5.5}{0.287 \times T_2} = \frac{22996.5}{T_2}\,kg$$

$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\gamma-1/\gamma} \Rightarrow T_2 = 315\left(\frac{12}{18}\right)^{.4/1.4} = 290\,K$

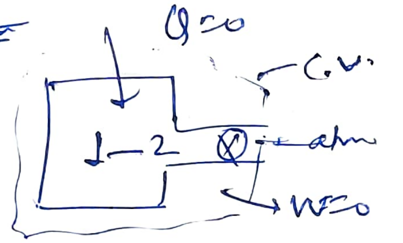
$m_2 = 79.3\,kg$

$m_e = m_1 - m_2 = 97.34 - 79.3\,kg = 18.04\,kg$

Q. $1.6\,m^3$ tank is filled with air at a pressure of 5 bar and a temperature of 100°C. The air is then let off to the atmosphere through a valve. Assuming no heat transfer, determine the work obtainable by utilising the kinetic energy of the discharge air to run a frictionless turbine.

Soln

~~atm~~




$V = ?$ $\left(\frac{m_e V^2}{2}\right) \rightarrow$ K.E.

Take- $C_p = 1$ kJ/kgK, $C_v = 0.711$ kJ/kgK atmp - 1 bar

$R = C_p - C_v = 0.289$ kJ/kgK

Soln

1 → 2

$V_1 = 1.6\,m^3$ | $V_2 = 1.6\,m^3$

$P_1 = 5\,bar$ | $P_2 = 1\,bar$

$T_1 = 100°C \rightarrow 373\,K$ | $T_2 = ?$

mass balance → $\cancel{m_i}^0 - m_e = m_2 - m_1$

$\Rightarrow m_e = m_1 - m_2$

Energy balance → $m_i h_i^0 + Q^0 = m_e\left(h_e + \frac{V_e^2}{2}\right) + W^0 + m_2 u_2 - m_1 u_1$

$\Rightarrow m_e\left(h_e + \frac{V_e^2}{2}\right) = m_1 u_1 - m_2 u_2$

$\Rightarrow m_e \frac{V_e^2}{2} = m_1 u_1 - m_2 u_2 - m_e h_e$
$= m_1 u_1 - m_2 u_2 - (m_1 - m_2) h_e$

$m_1 = \frac{P_1 V_1}{R T_1} = \frac{5\times10^2 \times 1.6}{0.289 \times 373} = 7.42\,kg$

$m_2 = \frac{P_2 V_2}{R T_2}$ . $\left(\frac{P_2}{P_1}\right)^{\gamma-1/\gamma} = T_2/T_1 \Rightarrow T_2 = \left(\frac{1}{5}\right)^{.4/1.4} T_1 = 0.63 \times 373 = 235.36\,K$

finally in the

$$m_x = \frac{P_2 V_2}{R T_2} = 2.35\,kg.$$

$$m_e \frac{V_e^2}{2} = (m_1 u_1 - m_2 u_2) - (m_1 - m_2) h_e$$

$$= (m_1 C_v T_1 - m_2 C_v T_2) - (m_1 - m_2) C_p T_2$$

$$= (7.42 \times 0.771 \times 373 - 2.35 \times 0.771 \times 235.36)$$

$$- (7.42 - 2.35) \times 1 \times 235.36$$

$$K.E. = 514.15\ KJ$$